

Selected Projects: PhD Discussion Status

Application-facing scientific-audit markings for deciding which portfolio projects to discuss in PhD applications and interviews.

STRONG TO DISCUSS	Audited; suitable for substantive discussion when the stated limitations are preserved.
DISCUSS WITH QUALIFICATION	Scientifically meaningful, but a correction, convergence issue, or pending verification must be disclosed.
DO NOT USE AS A HEADLINE YET	Important unresolved verification/scope issue; use only with explicit caveats when directly relevant.
AUDIT PENDING	Not yet reviewed under the same rigorous problem-method-implementation framework.

Current priority: for applications today, lead with the strongest audited projects that match the advertised PhD. A negative result is not a weakness when you can explain the mathematical reason, the diagnostic evidence, and what you would test next.

1. 2D Incompressible Navier-Stokes: Pseudo-Spectral Solver and POD-Galerkin ROM

DISCUSS WITH QUALIFICATION

The PDE and methods are appropriate, and the inconsistent 2/3-dealiasing/forcing defect has been corrected. In an interview, explain the correction openly. Treat the existing full-study ROM numbers as provisional until the corrected FOM/ROM chain is rerun. The ~99% unseen-realisation POD projection failure is a genuine negative result and is worth discussing.

2. Finite-Element Solver for 2D Incompressible Flow: Schafer-Turek Cylinder Benchmarks

STRONG TO DISCUSS

Right problem and right Taylor-Hood FEM method, with strong manufactured-solution verification. Present 2D-1 as well supported and 2D-2 as close to published benchmark values with residual force/post-processing sensitivity. Do not imply that 2D-3 was implemented.

3. Phase-Field Modelling of Boiling Heat Transfer: Toward 3D DNS

DO NOT USE AS A HEADLINE YET

The dedicated 1D Stefan solver and prescribed-rate verification are defensible, but the general 2D heat-flux pathway is not verified and 3D heat-flux boiling is not implemented. Discuss only when phase-change modelling is directly relevant, and state this scope explicitly.

4. Optimal Mixing of Passive Scalars: M.Sc. Thesis and Numerical Convergence Study

STRONG TO DISCUSS

A genuine thesis-level PDE/numerical-analysis project. The greedy LTD strategy is appropriate for instantaneous stirring optimisation. Emphasise the later convergence/resolution analysis and do not claim that the simulations establish a universal asymptotic a^{-1} scaling law.

5. Deterministic and Bayesian Inversion of Spatially Varying Darcy Permeability

DISCUSS WITH QUALIFICATION

The elliptic inverse problem, adjoint/Tikhonov method and pCN formulation are appropriate. Explain that sharp-inclusion recovery exposes prior/regulariser mismatch, the reported PDE chains have low effective sample sizes, and an optimiser endpoint that did not converge is a MAP candidate rather than an established MAP estimate.

6. Photoacoustic Reconstruction: Model-Based vs. Learned Inversion

STRONG TO DISCUSS

The inverse problem and model-based/learned methods are legitimate, with careful protocol controls. A particularly strong research point is that learned in-distribution gains can fail on thin vessel-like structures. Present that as a distribution-shift limitation rather than hiding it.

7. Ensemble Kalman Filtering for the Chaotic Lorenz-96 System

STRONG TO DISCUSS

The stochastic EnKF, cyclic localisation and inflation setup are appropriate and independently checked against a linear-Gaussian Kalman reference. Small-ensemble failure and under-dispersion are meaningful negative results. Keep conclusions specific to the tested perfect-model Lorenz-96 regimes.

8. 2D Elasticity FEM and SIMP Topology Optimisation

DISCUSS WITH QUALIFICATION

The elasticity/SIMP formulation and sensitivity verification are sound, but the raw-design versus filtered-physical-volume consistency in the OC feasibility check still needs targeted verification. Do not describe this as stress-constrained optimisation: von Mises stress is post-processing only.

9. ROM, Hyper-Reduction, and Neural Operators for Viscous Burgers' Equation

STRONG TO DISCUSS

POD-Galerkin, DEIM and FNO are appropriate comparison methods. The historical reduced-timestep issue is explicitly diagnosed and a revised stability policy exists. Discuss projection error, reduced dynamics, hyper-reduction and learned-operator generalisation as distinct questions rather than claiming one method is universally superior.

10. Physics-Informed Neural Network: 1D Advection-Diffusion

DISCUSS WITH QUALIFICATION

The PDE is valid and the current periodic formulation correctly enforces both value and derivative matching. The historical value-only periodic condition was incomplete. Present the PINN as a scientific-ML/optimisation study, not as a computationally superior solver for this simple linear PDE.

Projects not yet audited under the current framework

These are not being labelled incorrect. Their status is simply not yet established to the same standard as the first ten projects.

11. PhysicsNeMo Stokes Flow: Data-Driven MeshGraphNet and Physics-Informed Fine-Tuning

AUDIT PENDING

Not yet reviewed under the same problem-method-implementation audit. Do not rely on it as a primary PhD discussion project until checked.

12. PhysicsNeMo PINN Lid-Driven Cavity: Reynolds-Number Study

AUDIT PENDING

Not yet reviewed under the same problem-method-implementation audit. Do not rely on it as a primary PhD discussion project until checked.

13. PhysicsNeMo FNO for Darcy Flow: Operator-Learning Reproduction

AUDIT PENDING

Not yet reviewed under the same problem-method-implementation audit. Do not rely on it as a primary PhD discussion project until checked.

14. PhysicsNeMo PINO for Darcy Flow: Physics-Loss Weighting Study

AUDIT PENDING

Not yet reviewed under the same problem-method-implementation audit. Do not rely on it as a primary PhD discussion project until checked.

15. PhysicsNeMo MeshGraphNet for Vortex Flow: Capacity and Rollout Study

AUDIT PENDING

Not yet reviewed under the same problem-method-implementation audit. Do not rely on it as a primary PhD discussion project until checked.

16. Differentiable PBR: Inverse Material Optimisation

AUDIT PENDING

Not yet reviewed under the same rigorous scientific audit. It may be useful for inverse-problem/optimisation applications, but its claims should be checked before using it as a headline technical example.

17. TV Image Inpainting via Preconditioned Douglas-Rachford Splitting

AUDIT PENDING

Not yet reviewed under the same rigorous scientific audit. Check the optimisation formulation, convergence assumptions and numerical comparisons before relying on detailed claims.

18. 3D Liver Segmentation: Abdominal CT

AUDIT PENDING

Not yet reviewed under the same audit. The evaluation protocol has received prior attention, but this document does not yet certify the project's scientific claims as a whole.

19. Aircraft Engine Monitoring via Audio DSP

AUDIT PENDING

Not yet reviewed under the same audit. It may be useful for applied signal-processing/ML roles, but it is not yet classified for rigorous PhD discussion.

20. INTUOS Flight Data Monitoring Dashboard

AUDIT PENDING

Not yet reviewed under the same audit. Better suited to applied data/engineering discussion unless and until its scientific claims are separately audited.

21. NeRF vs TensorRF: Neural Radiance Fields

AUDIT PENDING

Not yet reviewed under the same audit. Do not rely on comparative performance or methodological claims until checked.

22. S4D vs. Transformer: A Controlled Sequence-Architecture Comparison

AUDIT PENDING

Not yet reviewed under this scientific audit. CUDA/provenance work exists, but the full research-question and comparison protocol still need the same systematic review.

23. LLM Engineering Portfolio: Applied Generative-AI Projects

AUDIT PENDING

Not yet reviewed under this scientific audit. Treat primarily as engineering evidence rather than a core numerical-PDE/scientific-ML research project.

Best current projects to lead with

Numerical PDE / CFD: FEM cylinder flow; M.Sc. optimal-mixing thesis. Navier-Stokes is also highly relevant, but until the corrected experiment chain is rerun, discuss it as a project where you identified and corrected a discrete dealiasing inconsistency.

Reduced-order modelling / operator learning: Burgers POD-Galerkin/DEIM/FNO. Navier-Stokes is valuable for discussing why a POD basis can be excellent in-sample yet fail geometrically on an unseen flow realisation.

Inverse problems / uncertainty: Photoacoustics is currently the safest headline. Darcy is also valuable if you explicitly discuss identifiability, prior misspecification, low ESS and the difference between a MAP candidate and a converged MAP estimate.

Data assimilation: Lorenz-96 EnKF is strong, especially if you discuss filter failure and under-dispersion rather than only successful runs.

Scientific ML / PINNs: the advection-diffusion PINN is discussable as a formulation and optimisation study, but it should not be presented as evidence that PINNs outperform classical solvers on this PDE.

For now: do not lead with the boiling project until the general heat-flux pathway is verified; do not make topology optimisation a headline until the filtered-volume consistency check is completed.